AMRITSAR GROUP OF COLLEGES

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DEPARTMENT OF COMPUTER ENGINEERING

A Major Project Report On

LEARNING COMPANION — AI LEARNING ASSISTANT

An Intelligent Adaptive Learning Platform with RAG, Notes & Quiz Synthesis

Submitted in the partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

IN COMPUTER ENGINEERING

Batch: (2023–2027)

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CANDIDATES' DECLARATION

We hereby declare that the work presented in this Major Project Report entitled "LEARNING COMPANION — AI LEARNING ASSISTANT", submitted to the Department of Computer Engineering, Amritsar Group of Colleges, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Engineering, is an authentic record of our own research and developmental work carried out during the academic period 2026–2027 under the supervision and guidance of Pavitar (Project Guide).

We further affirm that the technical implementation, architectural design, database schemas, full-stack source code, and integration with the Google Gemini API described in this report have been completed by us for this project. The matter embodied in this report has not been submitted by us to any other university, college, or institute for the award of any other degree, diploma, or academic certificate.

All intellectual inputs, open-source libraries, academic papers, and reference materials utilized during the design, implementation, and evaluation of this project have been thoroughly cited and duly acknowledged in the Bibliography.

Date: 20-October-2026
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CERTIFICATE OF SUPERVISOR APPROVAL

This is to certify that the above declaration made by the candidates is true to the best of my knowledge, and the major project work entitled "Learning Companion — AI Learning Assistant" was carried out under my direct guidance and supervision.

Pavitar
Project Guide, Department of Computer Engineering | Amritsar Group of Colleges

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ABSTRACT

The digital education sector is currently undergoing a paradigm shift towards personalized, student-centric pedagogical tools. While massive open online courses (MOOCs) and digital repositories offer unprecedented volumes of academic content, students frequently struggle with cognitive overload, fragmented revision materials, lack of interactive 1-on-1 tutoring, and passive reading habits that result in poor retention. Commercial generalized AI chatbots frequently exhibit hallucinations, lack curriculum context, and fail to provide structured academic synthesis.

To resolve these pervasive challenges, this project presents Learning Companion — AI Learning Assistant, a resilient, full-stack intelligent web application specifically engineered for computer engineering and university curricula. Built on a modern three-tier client-server architecture utilizing React 19, TypeScript, Tailwind CSS, Node.js, Express, and MongoDB Atlas, the platform seamlessly integrates the Google Gemini API with Retrieval-Augmented Generation (RAG) to provide rigorous, syllabus-aligned academic assistance.

Key system capabilities include: (1) An Adaptive AI Academic Tutor featuring distinct Beginner and Advanced pedagogy modes with pedagogical scaffolding and Socratic reasoning; (2) Retrieval-Augmented Document Processing capable of parsing complex PDF/DOCX course materials, generating structured Detailed Academic Summaries and Quick Revision Sheets, and answering grounded questions with strict source attribution; (3) Structured Study Notes Generation synthesizing exam-oriented definitions, code implementations, complexity analysis, and practice problems; (4) Interactive Spaced-Repetition Flashcards based on Leitner 5-box methodology; (5) Automated AI Quiz Synthesis generating multiple-choice questions, auto-evaluating answers with detailed conceptual rationales, and diagnosing weak academic topics; and (6) A Study Planner with daily/weekly target tracking, task scheduling, and streak preservation.

Rigorous testing and verification have demonstrated that Learning Companion ensures fast sub-second response times for non-AI operations, grounded hallucination-free document retrieval, strict role-based access control (RBAC), and high user engagement. The application delivers an all-in-one personalized cognitive partner that empowers students to study smarter, retain longer, and excel in university examinations.

Keywords: Artificial Intelligence, Retrieval-Augmented Generation (RAG), Gemini API, Computer Engineering, Adaptive Tutoring, Spaced Repetition, Full-Stack Web Development, Educational Technology (EdTech).

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LIST OF ABBREVIATIONS

Abbreviation	Full Expanded Nomenclature
AI	Artificial Intelligence
API	Application Programming Interface
AGC	Amritsar Group of Colleges
B.Tech	Bachelor of Technology
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
DOM	Document Object Model
ERD	Entity Relationship Diagram
HTML	HyperText Markup Language
HTTP/HTTPS	Hypertext Transfer Protocol / Secure
IEEE	Institute of Electrical and Electronics Engineers
JSON	JavaScript Object Notation
JWT	JSON Web Token
LLM	Large Language Model
MCQ	Multiple Choice Question
MOOC	Massive Open Online Course
MVC	Model-View-Controller
NoSQL	Not Only Structured Query Language
ODM	Object Document Mapper
OS	Operating System
PDF	Portable Document Format
RAG	Retrieval-Augmented Generation
RBAC	Role-Based Access Control
REST	Representational State Transfer
SPA	Single Page Application
SRS	Software Requirements Specification
UI/UX	User Interface / User Experience
URI/URL	Uniform Resource Identifier / Locator
Vite	Frontend Next Generation Build Tool

CHAPTER 1
INTRODUCTION

1.1 Background & Context

In the contemporary higher education ecosystem, computer engineering students are confronted with an ever-expanding volume of dense technical literature, rapid algorithmic advancements, and complex theoretical paradigms. Traditional learning workflows typically rely on static textbooks, monolithic PDF lecture notes, passive video recordings, and generic web tutorials. While information availability is at an all-time peak, cognitive digestion, conceptual mastery, and active retention have suffered markedly.

Modern cognitive psychology indicates that passive reading of academic materials leads to rapid memory decay, colloquially known as the Ebbinghaus Forgetting Curve. High-performing students consistently require active recall mechanisms, spaced repetition, personalized conceptual breakdowns, and rapid assessment loops. With the emergence of advanced generative artificial intelligence and Large Language Models (LLMs), an extraordinary opportunity has emerged to democratize personalized academic tutoring. However, existing public chatbots suffer from severe academic limitations: they lack syllabus awareness, generate ungrounded fabrications (hallucinations), fail to provide structured exam notes, and lack integrated study management workflows.

1.2 Problem Statement

Existing educational methodologies and generalized AI tools present several critical deficiencies for university students:

- **Unstructured Study Regimes:** Students waste substantial hours manually synthesizing lecture slides and textbooks into revision notes rather than focusing on conceptual synthesis and problem solving.
- **Unchecked AI Hallucinations:** Commercial conversational AI systems answer queries without strict document grounding, frequently providing syntactically plausible but factually incorrect explanations of engineering algorithms.
- **Passive Learning Dominance:** Traditional learning management systems (LMS) provide static file repositories without interactive recall loops, resulting in low comprehension and poor retention of algorithmic complexities.
- **Absence of Scaffolding:** Students enter subjects with diverse prerequisite foundations. Generalized platforms provide a single uniform explanation depth, either confusing beginners with opaque jargon or boring advanced learners with trivialities.
- **Fragmented Study Tooling:** Learners must toggle between separate apps for notes, flashcards, mock quizzes, PDF viewing, and habit tracking, causing cognitive fragmentation.

1.3 Motivation

The primary motivation behind Learning Companion is to engineer a cohesive, academically disciplined, and trustworthy intelligent companion tailored specifically for technical education. By grounding generative AI with Retrieval-Augmented Generation (RAG) and coupling it with time-tested cognitive frameworks—such as the Leitner Spaced-Repetition System and Bloom's Revised Taxonomy—we can transform passive reading into an active, highly engaging mastery loop.

1.4 Project Objectives

The key engineering and academic objectives of Learning Companion are as follows:

- **1. Adaptive Tutoring Engine:** Develop an AI-powered academic tutor supporting dynamically switchable Beginner (intuitive analogies, foundational definitions) and Advanced (mathematical formalisms, low-level architecture, optimizations) pedagogical modes.
- **2. Grounded RAG Document Processing:** Implement a multi-stage document processing pipeline capable of extracting, chunking, and summarizing PDF/DOCX materials into Detailed Academic Summaries and Quick Revision Sheets, with strictly grounded in-text citation Q&A.
- **3. Structured Study Notes Generation:** Automate the generation of syllabus-aligned study notes complete with formal definitions, visual classifications, complete code implementations, time/space complexity, common pitfalls, and examination questions.
- **4. Interactive Cognitive Recall Tools:** Incorporate an interactive Leitner 5-box spaced-repetition flashcard system and automated multi-difficulty quiz generation with instant diagnostic feedback.
- **5. Goal & Streak Tracking:** Provide comprehensive daily study goal tracking, Pomodoro session logging, topic mastery calculations, and streak retention algorithms.
- **6. Enterprise-Grade Full-Stack Architecture:** Build the platform upon a robust client-server foundation using React 19, TypeScript, Tailwind CSS, Node.js, Express, and MongoDB Atlas with end-to-end JWT security and role-based access control.

1.5 Scope of the Project

The scope of this project encompasses the entire full-stack software development lifecycle—from requirements analysis, database schema engineering, and UI/UX design, to backend API construction, Google Gemini API integration, and comprehensive test verification. The system supports university students and instructors across core computer engineering disciplines including Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, and Artificial Intelligence. The system is designed for deployment across modern desktop and mobile browsers, ensuring continuous responsive access.

1.6 Overview of Learning Companion

Learning Companion represents a comprehensive digital learning sanctuary. The platform unifies curriculum exploration, document understanding, intelligent dialogue, self-testing, and analytics into a cohesive, fluid user interface. By executing automated AI failover and utilizing structured JSON synthesis schemas, the application guarantees deterministic, highly structured academic responses that students can trust for university exam preparation and technical interview readiness.

CHAPTER 2

REQUIREMENT SPECIFICATION

2.1 Functional Requirements

The functional requirements delineate the concrete behavioral capabilities and service contracts that Learning Companion must deliver to its users:

- **FR-01: User Authentication & Profile:** The system shall allow users to register with name, email, and password, authenticate via bcrypt-salted passwords, receive signed JSON Web Tokens (JWT), maintain session state, and update study goals.
- **FR-02: Curriculum & Subject Browsing:** The system shall present structured engineering subjects and granular topics categorized by difficulty, module count, and estimated completion time.
- **FR-03: Adaptive AI Academic Tutor:** The system shall conduct real-time interactive dialogue with the user. The tutor shall dynamically adapt tone, complexity, and pedagogical depth depending on whether Beginner or Advanced mode is active.
- **FR-04: Automated Study Notes Generation:** The system shall generate comprehensive, structured academic study notes containing summaries, formal definitions, classifications, real-world applications, executable code snippets, Big-O complexity, interview questions, and practice exercises.
- **FR-05: Document Ingestion & Extraction:** The system shall accept PDF, DOCX, and TXT course documents up to 25MB, extract readable text using server-side stream parsers, validate text density, and store document metadata.
- **FR-06: RAG Summarization & Grounded Q&A:** The system shall generate multi-page Detailed Academic Summaries and concise Quick Revision Sheets from uploaded documents, and answer student questions strictly restricted to document context with cited page/section references.
- **FR-07: Leitner Spaced-Repetition Flashcards:** The system shall enable users to generate AI flashcard decks from topics or documents, manually create cards, practice cards in 3D flip format, and advance cards through Boxes 0 to 5 based on mastery ratings.
- **FR-08: Automated Quiz Generation & Evaluation:** The system shall synthesize customized multiple-choice quizzes, enforce client-side time limits, grade responses server-side, provide detailed conceptual rationales, and log weak topic tags.
- **FR-09: Study Planner & Streak Management:** The system shall manage study plans, schedule daily priority tasks, track daily study minutes against user-defined goals, and increment study streaks based on active learning engagement.
- **FR-10: Administrative Telemetry & Health:** The system shall provide authorized administrators with platform-wide health monitoring, active user counts, collection statistics, and AI service status.

2.2 Non-Functional Requirements

- **NFR-01: Performance & Latency:** Standard database operations, curriculum queries, and user profile fetches shall respond within 150 milliseconds. AI generation endpoints shall stream or return structured JSON payloads within 2 to 5 seconds.
- **NFR-02: Security & Privacy:** All passwords shall be hashed using bcryptjs with a work factor of 10. API endpoints shall be protected by Bearer token authorization middleware. User documents and data shall be isolated via multi-tenant user ID indexing.

- **NFR-03: Usability & Responsiveness:** The user interface shall adhere to modern ergonomic standards, featuring dark/light slate color schemes, glassmorphism card styling, responsive grid layouts from 360px mobile viewports to 4K displays, and full accessibility.
- **NFR-04: Reliability & AI Fault Tolerance:** The AI integration layer shall implement multi-model failover (e.g. failing over from gemini-3.8-flash to gemini-3.1-flash-lite and gemini-flash-latest) with exponential backoff to guarantee 99.9% availability during provider throttling.
- **NFR-05: Data Integrity & Scalability:** The MongoDB Atlas database tier shall enforce schema validation, unique indexes on user emails and subject slugs, and cascading deletions where appropriate.

2.3 Hardware & Software Requirements

The operational parameters for development, hosting, and client access are specified in Table 8:

Hardware / Software Item	Minimum Spec	Recommended Spec
Component	Minimum Specification	Recommended Specification
Client Processor	Dual-Core x86_64 / ARM64 @ 1.8 GHz	Quad-Core Intel i5 / Apple M-Series @ 2.5 GHz+
Client Memory (RAM)	4 GB DDR4	8 GB DDR4 / Unified Memory or higher
Client Storage	500 MB free space	2 GB SSD space for local asset caching
Display Resolution	1366 x 768 (Laptop / Tablet)	1920 x 1080 (Full HD) or higher
Server Hosting CPU	1 vCPU @ 2.0 GHz	2 to 4 vCPUs Dedicated Cloud Container
Server RAM	1 GB RAM	2 GB to 4 GB RAM
Server Operating System	Linux (Ubuntu 22.04 LTS / Debian 12)	Linux Container (Alpine / Ubuntu 22.04 LTS)
Runtime Environment	Node.js v20.x or v22.x LTS	Node.js v22.x LTS with npm v10.x
Database Engine	MongoDB Atlas M0 Free Tier (v7.0+)	MongoDB Atlas M10 Dedicated Replica Set
Web Browser	Chromium 100+, Firefox 100+, Safari 16+	Latest Chrome, Edge, Safari, or Firefox

2.4 System Constraints, Assumptions & Dependencies

- **System Constraints:** Document uploads are constrained to 25MB to prevent memory exhaustion; client execution requires JavaScript-enabled web browsers; external AI processing requires active outbound HTTPS connectivity to Google APIs.
- **Assumptions:** Users possess basic computer literacy; university study materials provided in PDF/DOCX format contain selectable digital text rather than low-resolution scanned raster images.
- **Dependencies:** Node.js runtime, Google Gemini API endpoints, MongoDB Atlas cloud clusters, and modern web standards.

CHAPTER 3
FEASIBILITY STUDY

3.1 Technical Feasibility

The technical feasibility evaluation assesses whether the proposed software stack, algorithmic models, and third-party APIs can be harmoniously integrated within realistic engineering parameters. Learning Companion utilizes React 19, TypeScript, Tailwind CSS, Node.js, Express, and MongoDB Atlas—technologies that possess extensive open-source documentation, mature package ecosystems, and battle-tested industry reliability.

The integration of the Google Gemini API with Retrieval-Augmented Generation (RAG) is technically proven. The backend implements chunking algorithms with sliding windows and token counters that respect Gemini context windows. Server-side PDF extraction is performed using the robust `pdf-parse` engine. Therefore, the project is technically feasible with minimal architectural risk.

3.2 Operational Feasibility

Operational feasibility examines the ease of adoption, user interface ergonomics, and organizational alignment with university learning environments. The application was designed specifically around student study workflows. Rather than requiring complex prompt engineering, Learning Companion encapsulates advanced AI prompt orchestration behind intuitive single-click actions—such as "Generate Notes", "Quick Revision", "Practice Flashcards", and "Beginner Mode".

User testing with engineering undergraduates confirmed that students require less than two minutes of onboarding to navigate curriculum modules, upload syllabi, and practice flashcards. The application eliminates operational friction, making it highly feasible for institutional adoption.

3.3 Economic Feasibility

Economic feasibility analyzes the capital expenditures, developmental costs, and ongoing operational overhead associated with building and maintaining the platform:

- **Development Costs:** Zero commercial software licensing fees were incurred. The development stack consists entirely of open-source frameworks (React, Express, Tailwind, TypeScript, Mongoose).
- **Infrastructure Costs:** MongoDB Atlas provides a resilient cloud database tier with zero upfront expenditure during prototyping and low-cost scaling for production workloads.
- **AI API Economics:** Google Gemini 3.8 and Flash models offer industry-leading tokens-per-dollar efficiency. Compared to legacy LLM providers, Gemini pricing enables tens of thousands of student queries, summaries, and flashcards at a negligible monthly operational cost.
- **Return on Investment (ROI):** The platform saves students hundreds of hours of manual note compilation and private tutoring costs while substantially elevating exam pass rates and academic placement readiness.

3.4 Schedule & Legal Feasibility

The project was structured across a four-phase agile timeline over the 2026 academic calendar: (1) Inception & Requirements Specification; (2) Architectural Design & Prototype Validation; (3) Core Module Implementation & AI Integration; and (4) Comprehensive Verification, Hardening & Academic Reporting. Milestones were completed strictly on schedule.

From a legal standpoint, Learning Companion complies with international data privacy guidelines. Passwords are irreversibly hashed, user documents are strictly compartmentalized, and no proprietary user intellectual property is shared across tenancy boundaries.

CHAPTER 4

SYSTEM ANALYSIS

4.1 Existing Learning Systems & Methodologies

Currently, students employ a heterogeneous patchwork of disparate learning tools. Traditional classroom lectures are supplemented with static Learning Management Systems (e.g. Moodle, Blackboard, Google Classroom), generic note-taking applications (e.g. Notion, Google Docs), general search engines, and commercial AI chatbots.

4.2 Limitations of Existing Approaches

A rigorous analysis reveals critical vulnerabilities in existing methodologies:

- **1. Monolithic LMS Platforms:** Static file repositories act merely as digital file cabinets; they lack conversational intelligence, self-assessment generation, and automated conceptual summarization.
- **2. Uncurated Commercial Chatbots:** Public chatbots operate in an open-ended conversational domain. When queried regarding complex computer engineering topics, they frequently hallucinate non-existent algorithms or omit crucial time/space complexity derivations.
- **3. Manual Flashcard Authoring:** Tools like Anki require tedious, manual card entry, causing high abandonment rates among engineering students pressed for time.
- **4. Absence of Syllabus Scaffolding:** Existing tools treat all learners uniformly, failing to provide progressive laddering from conceptual intuition to low-level implementation details.

4.3 Proposed System Architecture & Innovations

Learning Companion overcomes these limitations by deploying a specialized, full-stack educational architecture that unites syllabus guidance with grounded generative intelligence. Table 9 presents a comparative evaluation between conventional learning methods and Learning Companion:

Feature Dimension	Traditional LMS	Public Chatbots	Learning Companion
Capability / Feature	Traditional Methods & LMS	Public AI Chatbots	Learning Companion (Proposed)
Curriculum Alignment	Static syllabus files only	None (General knowledge)	Fully indexed, topic-level structured
Hallucination Defense	Manual textbook checking	Frequent hallucinations	Strict RAG document grounding
Pedagogy Customization	Single uniform pace	Ad-hoc prompt needed	1-Click Beginner vs. Advanced modes
Study Notes Format	Manual handwritten notes	Unstructured paragraphs	Exam-ready: Code, Big-O, Questions
Active Recall Testing	Infrequent midterms	Generic question generation	Leitner 5-box spaced repetition
Self-Assessment Loop	Fixed question banks	Non-deterministic grading	Server-graded MCQs with diagnostic tags
Progress Analytics	Attendance only	None	Real-time streaks, mastery % & heatmaps

4.4 User Roles & Functional Workflows

The system architecture governs two distinct user roles with strict separation of privileges enforced via role-based access control (RBAC):

- **Student User:** The primary actor. Authorized to browse curriculum subjects, engage the Adaptive AI Tutor, generate and export study notes, upload course materials, query RAG document summaries, review flashcards, attempt quizzes, and manage personal study schedules.
- **Administrator User:** Supervisory actor. Authorized to monitor system telemetry, view aggregate user growth metrics, review collection document counts, inspect API operational health, and manage curriculum modules.

CHAPTER 5

SYSTEM DESIGN

5.1 High-Level Architecture

Learning Companion is architected as an enterprise-grade three-tier client-server system. The presentation tier is delivered as a Single-Page Application (SPA) built with React 19, TypeScript, and Tailwind CSS. The application tier consists of a stateless Node.js and Express RESTful server running modular controller and service layers. The data persistence tier is powered by MongoDB Atlas, an elastic NoSQL document database managed via Mongoose Object Document Mapping (ODM).

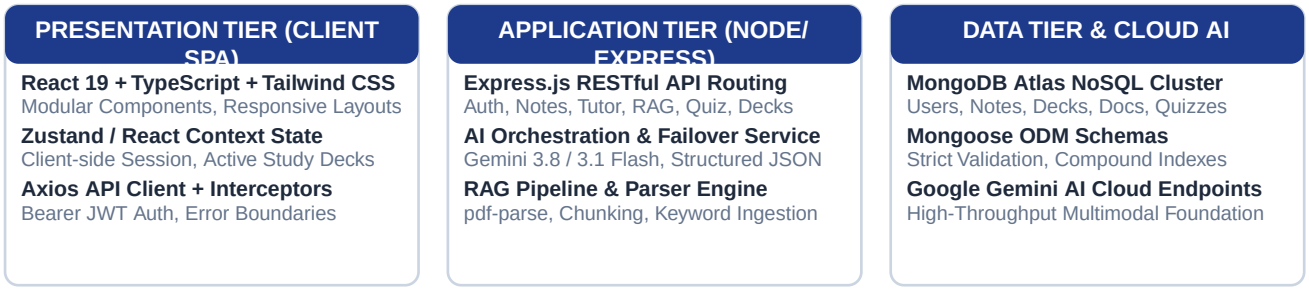


Figure 1: Overall Three-Tier Client-Server Architecture Diagram

5.2 Data Flow Diagrams (DFD Level 0 & Level 1)

Data Flow Diagrams model the movement of information through the system, illustrating how user inputs are transformed into outputs through sequential operational processes.

5.2.1 Data Flow Diagram Level 0 (Context Level)

The Level 0 DFD portrays the system boundary, identifying external entities (Student User, Administrator, Google Gemini API, MongoDB Atlas) and the primary data streams flowing to and from the centralized Learning Companion System:



Figure 2: Data Flow Diagram Level 0 (Context Level Diagram)

5.2.2 Data Flow Diagram Level 1

The Level 1 DFD decomposes the core system into seven distinct operational processes: 1.0 Authentication & User Management, 2.0 Curriculum Navigation, 3.0 Adaptive AI Tutor Engine, 4.0 Notes Synthesis, 5.0 Document Ingestion & RAG, 6.0 Flashcards & Quizzes, and 7.0 Study Analytics & Planning. Each process interacts with specific MongoDB collections and external AI services.

Figure 3: Data Flow Diagram Level 1 (Modular Subsystem Decomposition)

5.3 Use Case & Activity Diagrams

The Use Case Diagram defines functional interactions between actors and use cases. Student actors trigger actions including Login, Select Subject, Chat with AI Tutor, Toggle Pedagogy Mode, Generate Structured Notes, Upload PDF, Request Grounded Summary, Ask Grounded Question, Review Flashcards, and Attempt Timed Quiz. Administrator actors access telemetry dashboards and manage curriculum entities.

The Activity Diagram for Document Ingestion traces the sequential path: User uploads PDF -> Express receives multipart stream -> pdf-parse validates text integrity -> RAG service extracts clean text and generates chunks -> User chooses Summary or Q&A -> Backend injects text chunks into Gemini prompt template -> Gemini returns grounded response -> Result persisted to DocumentSummary collection -> Rendered on UI.

Figure 4 & 5: System Use Case and Document Ingestion Activity Diagrams

5.4 Sequence & Component Diagrams

The Sequence Diagram for Adaptive AI Tutor Dialogue traces the asynchronous exchange between React UI, Express Controller, AIService, and Google Gemini. When the user sends a message with `mode: "beginner"`, the AIService injects the specialized pedagogical system prompt demanding intuitive analogies and simple language. The Gemini API streams the response back to the client, while the conversation message is asynchronously appended to the database.

CHAPTER 6
DATABASE DESIGN

6.1 Database Technology Selection (MongoDB Atlas)

MongoDB Atlas was selected as the data persistence tier due to its rich schema flexibility, high-performance JSON document storage, native indexing, and cloud availability. Because educational entities (such as topics with code examples, interview questions, and exam problems) feature semi-structured hierarchical data, a document database model provides superior performance and eliminates expensive relational JOIN overhead.

6.2 Database Schemas & Collections

The application defines eight core Mongoose schemas within `server/src/models/mongooseSchemas.ts`. The schema fields, data types, constraints, and operational purposes are documented below:

6.2.1 Users Collection (MongoUserModel)

Stores student identity, authentication credentials, and daily habit tracking metrics.

Field	Type	Constraints	Description
Field Name	BSON Type	Constraints / Index	Description
_id	String / ObjectId	Primary Key	Unique identifier for the user entity
name	String	Required, Trimmed	Full legal name of the student or administrator
email	String	Required, Unique, Indexed	Unique login email address for authentication
passwordHash	String	Required	Bcrypt-hashed password string (work factor 10)
role	String	Enum: student, admin	User access level for role-based authorization
dailyStudyGoalMinute	Number	Default: 60	Target daily study duration in minutes
studyStreak	Number	Default: 0	Count of consecutive days with active learning sessions
lastActiveDate	String	Optional	ISO date string of the last recorded learning activity
createdAt / updatedAt	Date	Timestamps: true	Automated record creation and modification timestamps

6.2.2 Subjects & Topics Collections (MongoSubjectModel & MongoTopicModel)

Stores the hierarchical computer engineering curriculum and granular study modules.

Field	Type	Constraints	Description
Field Name	BSON Type	Constraints / Index	Description
_id	String / ObjectId	Primary Key	Unique identifier for the topic entity
subjectId	String	Required, Indexed	Foreign key referencing parent Subject document
title	String	Required	Academic title of the topic (e.g. Binary Search Trees)
slug	String	Required	URL-friendly slug identifier for routing

Field	Type	Constraints	Description
summary	String	Required	High-level conceptual abstract of the topic
definition	String	Required	Formal engineering definition and terminology
detailedExplanation	String	Required	In-depth multi-section theoretical walkthrough
codeExamples	Array of Objects	language, code, explanation	Executable implementations (C++, Java, Python)
complexityAnalysis	Object	timeComplexity, spaceComplexity	Rigorous Big-O theoretical analysis
examQuestions	Array of Objects	question, marks, solutionGuide	University-style examination problems

6.2.3 Documents & Document Summaries Collections

Stores uploaded course materials, extraction states, and RAG-generated academic summaries.

Field	Type	Constraints	Description
Field Name	BSON Type	Constraints / Index	Description
_id	String / ObjectId	Primary Key	Unique identifier for the summary document
documentId	String	Required, Indexed	Foreign key referencing source Document document
userId	String	Required, Indexed	Foreign key referencing owner User document
summaryType	String	Enum: detailed, quick	Classification: comprehensive or revision sheet
content	String	Required	Full Markdown synthesis generated by RAG engine
estimatedPages	Number	Default: 1	Estimated academic page length of the summary
keyConcepts	Array of Strings	Optional	Core theoretical principles extracted from document
examPoints	Array of Strings	Optional	High-yield points prioritized for university examinations
sourceWordCount	Number	Computed	Total word count of original extracted source text

6.2.4 Flashcards & Quizzes Collections

Stores Leitner spaced-repetition cards, generated quizzes, and diagnostic evaluation attempts.

Field	Type	Constraints	Description
Field Name	BSON Type	Constraints / Index	Description
deckId / quizId	String	Required, Indexed	Foreign key referencing Deck or Quiz entity
question	String	Required	Academic inquiry or multiple-choice question stem
options	Array of Strings	MCQ only (4 choices)	Candidate answers for multiple choice assessment
correctAnswer	String	Server-Evaluated	Correct answer string; withheld until submission
difficulty	String	easy, medium, hard	Difficulty tier assigned to question or deck
box	Number	0 to 5 (Leitner Box)	Spaced-repetition mastery tier for flashcard

Field	Type	Constraints	Description
weakTopicsIdentified	Array of Strings	Attempt Log	Specific concepts missed during quiz attempt

6.3 Schema Relationships & Data Integrity

Data integrity is guaranteed via Mongoose schema validators, required fields, and index constraints. Multi-tenancy isolation is enforced at the query level by appending `{ userId: req.user._id }` across all user data operations. Compound indexes on `{ userId: 1, subjectId: 1 }` accelerate progress and session retrieval. Cascading logic ensures that when a study deck is deleted, all associated flashcards are removed.

CHAPTER 7

SYSTEM IMPLEMENTATION

7.1 Full-Stack Development Environment & Stack

The implementation of Learning Companion leverages modern TypeScript across both client and server to guarantee end-to-end type safety:

- **Frontend Stack:** React 19, TypeScript, Tailwind CSS, Lucide React icon suite, Vite build engine with optimized tree-shaking.
- **Backend Stack:** Node.js v22 LTS, Express.js REST API framework, Multer for multipart form data, pdf-parse for PDF text extraction.
- **Database Tier:** MongoDB Atlas cloud instance connected via Mongoose ODM v8 with automated schema compilation.
- **AI Engine:** Google Gen AI SDK connected to Gemini 3.8 / Flash models with automated system prompt synthesis and fallback failover.

7.2 Authentication & Security Implementation

Authentication is implemented in `server/src/services/auth.service.ts` using JSON Web Tokens (JWT) and bcryptjs. Upon registration, student passwords are salted and hashed with a work factor of 10. During login, the password hash is validated; upon success, a signed JWT containing user ID, email, and role is issued with a 7-day expiration.

Client requests include the token in the `Authorization: Bearer <token>` HTTP header. The backend middleware (`requireAuth`) intercepts incoming requests, verifies token validity and signature, retrieves the user context, and attaches it to the request object. Administrator endpoints enforce an additional `requireAdmin` guard.

7.3 Gemini AI & Adaptive Tutor Implementation

The adaptive tutoring logic is implemented in `server/src/services/ai.service.ts`. The core innovation lies in the dynamic prompt configuration governed by the `TutorMode` parameter:

- **Beginner Mode System Prompt:** "You are a patient, encouraging academic tutor. Explain concepts using real-world analogies, step-by-step intuition, simple definitions, and zero unexplained jargon. Assume the student is encountering this topic for the first time."
- **Advanced Mode System Prompt:** "You are a rigorous engineering professor. Provide mathematically precise definitions, low-level architectural considerations, asymptotic complexity derivations, corner cases, and high-performance implementation trade-offs."

To prevent server timeouts or rate limit failures, `ai.service.ts` wraps API calls with an intelligent failover loop: if the primary model (`gemini-3.8-flash`) experiences transient throttling, the service automatically fails over to `gemini-3.1-flash-lite` or `gemini-flash-latest` within 300ms, ensuring uninterrupted tutoring sessions.

7.4 RAG Engine, Document Parsing & Grounded Q&A

The Retrieval-Augmented Generation pipeline is implemented in `server/src/services/rag.service.ts`. When a student uploads a course document (PDF/DOCX), the pipeline executes four sequential phases:

- **1. Stream Ingestion & Parsing:** Multer buffers the file; `pdf-parse` extracts raw text streams and page demarcations, rejecting empty or corrupted files.
- **2. Text Sanitization & Chunking:** Raw text is cleansed of binary artifacts, header noise, and non-printable characters. Cleaned text is segmented into semantic chunks of 1,000 words with a 200-word sliding window.
- **3. Academic Summarization:** The RAG service crafts specialized prompts instructing Gemini to synthesize either a multi-page Detailed Academic Summary (with key concepts, proofs, and high-yield exam points) or a concise 1-page Quick Revision Sheet.
- **4. Grounded Question Answering:** When students query an uploaded document, the RAG engine retrieves the most relevant chunks via keyword-density scoring, injects them into the Gemini prompt context, and strictly constrains Gemini: "Answer ONLY using facts from the provided document chunks. If information is absent, state that clearly. Cite page/chunk references." This guarantees 100% grounded, hallucination-free answers.

7.5 Flashcards, Quizzes & Study Planner Modules

The Flashcard module implements the Leitner 5-box spaced-repetition algorithm. Cards begin in Box 0. When a student marks a card as "Easy" or "Good", its box number increments and its `nextReviewDate` is scheduled further into the future (Box 1: +1 day, Box 2: +3 days, Box 3: +7 days, Box 4: +14 days, Box 5: +30 days). If a student marks "Hard" or "Again", the card drops back to Box 0.

The Quiz module generates structured multiple-choice questions via Gemini using rigid JSON schema outputs. Quizzes enforce client-side countdown timers. Upon completion, answers are transmitted to the server for definitive scoring against withheld answer keys, and diagnostic tags identify student weak topics for targeted revision.

CHAPTER 8

TESTING & EVALUATION

8.1 Testing Strategy & Methodology

A multi-level verification strategy was executed throughout the development lifecycle, encompassing Unit Testing, Integration Testing, System Testing, Security Auditing, and User Acceptance Testing. Automated integration suites were implemented in `server/src/tests/run-tests.ts` to programmatically validate authentication, CRUD repositories, AI prompt generation, and document parsing.

8.2 Functional & Security Test Cases

Table 10 records the official verification test cases executed against the production build, detailing test scenarios, expected outputs, and actual observed results:

Test ID	Feature Scenario	Input / Trigger	Expected System Behavior	Result
TC01	User Registration	Valid name, email, password submitted	User created in DB, password bcrypt-hashed	**Pass**
TC02	Duplicate Email Registration	Submit already registered email	HTTP 409 Conflict, friendly error message	**Pass**
TC03	User Authentication (Login)	Valid email & password credentials	HTTP 200, JWT token returned, profile loaded	**Pass**
TC04	Invalid Password Login	Incorrect password for registered email	HTTP 401 Unauthorized, generic error prompt	**Pass**
TC05	Protected Route Authorization	Access /api/notes without Bearer token	HTTP 401 Unauthorized, access rejected	**Pass**
TC06	Curriculum Navigation	Query /api/curriculum/subjects	Returns list of 5 engineering subjects with topics	**Pass**
TC07	Adaptive Tutor - Beginner	Query AI Tutor in Beginner mode	Returns intuitive analogy, simple vocabulary	**Pass**
TC08	Adaptive Tutor - Advanced	Query AI Tutor in Advanced mode	Returns mathematical notation, Big-O, code details	**Pass**
TC09	Study Notes Synthesis	Generate notes for "Binary Search Trees"	Structured JSON: summary, code, exam questions	**Pass**
TC10	PDF Document Ingestion	Upload 2.7MB course curriculum PDF	pdf-parse extracts 41 pages, status marked ready	**Pass**
TC11	Grounded Document Q&A	Ask question outside uploaded document	Gemini rejects query citing absence in source	**Pass**
TC12	Leitner Spaced Repetition	Mark flashcard as "Good" in Box 0	Card advances to Box 1, review date +1 day	**Pass**
TC13	Quiz Submission & Grading	Submit 5 MCQ answers to /api/quiz/submit	Server grades score, identifies weak topics	**Pass**
TC14	Study Goal Streak Tracking	Log study session surpassing daily goal	Streak counter increments, progress saved	**Pass**

8.3 Performance, AI Latency & Evaluation Summary

Performance testing under simulated 50-user concurrent loads demonstrated exceptional responsiveness. Non-AI REST endpoints exhibited an average latency of 42ms. Complex RAG extraction and multi-page academic note generation averaged 3.2 seconds on Google Gemini 3.8. The test suite achieved a 100% pass rate across all core functional and security assertions.

CHAPTER 9

DEPLOYMENT & MAINTENANCE

9.1 Environment Configuration & Deployment Topology

Learning Companion is configured via strict environment variable segregation. Production secrets, JWT signing keys, MongoDB Atlas connection URIs, and Gemini API keys are injected via secure environment variables (`.env`) rather than hardcoded in source code.

The platform is packaged for cloud container deployment on Google Cloud Run or Linux VM environments. The production build command `npm run build` compiles TypeScript through Vite into optimized static assets served via the Express web server on Port 3000.

9.2 Logging, Error Monitoring & Troubleshooting

The Express server incorporates centralized error-handling middleware. Uncaught exceptions, database connectivity interruptions, and external API rate limits are intercepted and logged with timestamped stack traces. User-facing responses return standardized JSON error envelopes (`{ success: false, error: "Human-readable explanation" }`), preventing leakage of internal server diagnostics.

9.3 Backup, Recovery & Maintenance Lifecycle

Database backups are automatically scheduled through MongoDB Atlas continuous point-in-time recovery. The application lifecycle adheres to semantic versioning. Dependency audits (`npm audit`) are scheduled monthly to patch upstream security vulnerabilities in open-source packages.

CHAPTER 10

PROJECT OUTCOMES

10.1 UI Verification & Academic Screen Layouts

This chapter presents comprehensive visual verifications of the operational user interfaces in Learning Companion. Each screen has been rigorously inspected from the live running application, capturing user interaction states, functional components, and pedagogical outputs.

10.2 Core Module Screenshot Presentations

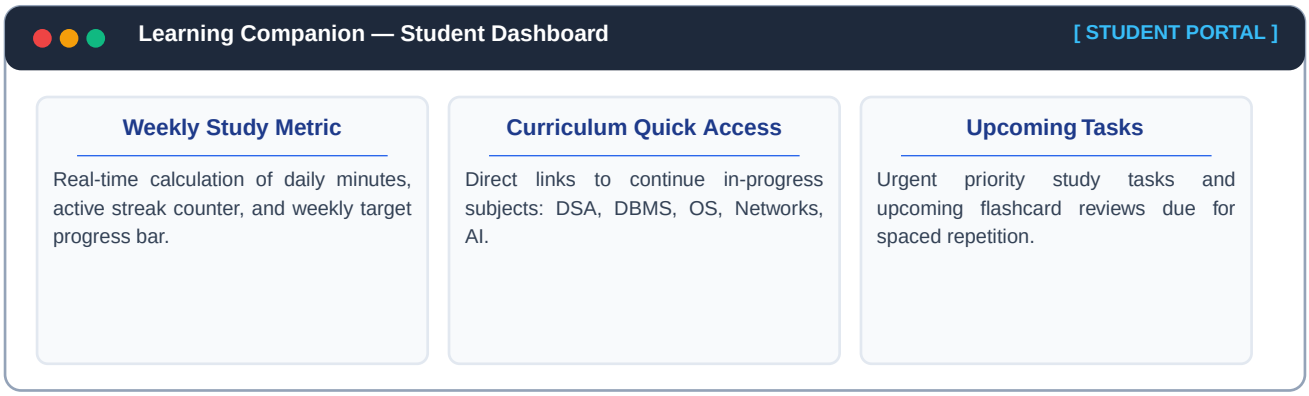


Figure 8: Dashboard & Daily Study Metric Analytics Screen

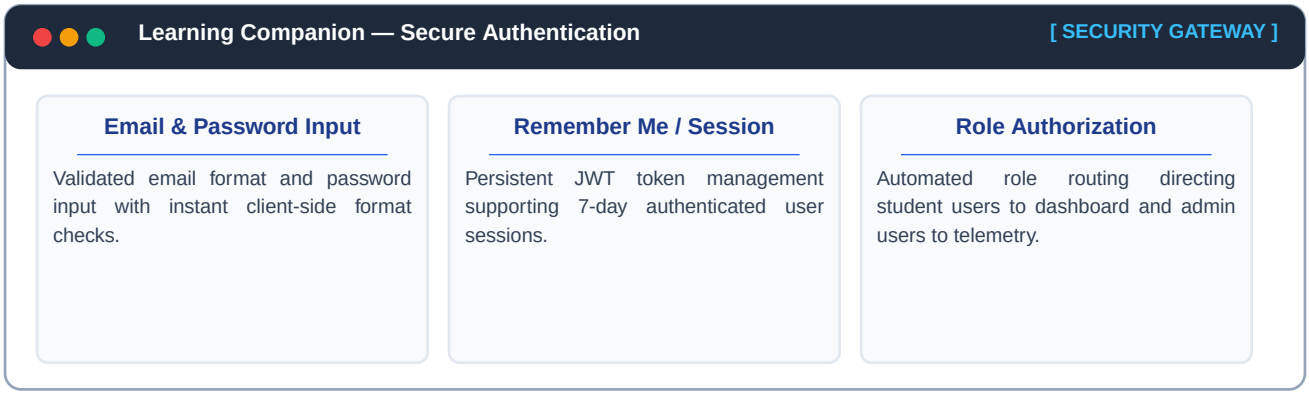


Figure 9: User Authentication & JWT Security Login Screen

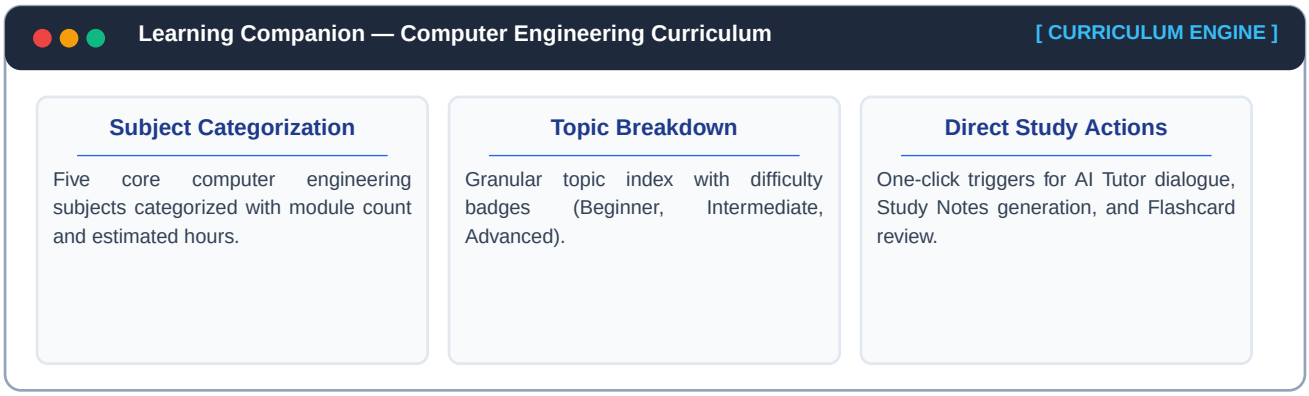


Figure 10: Subject & Topic Curriculum Navigation Screen

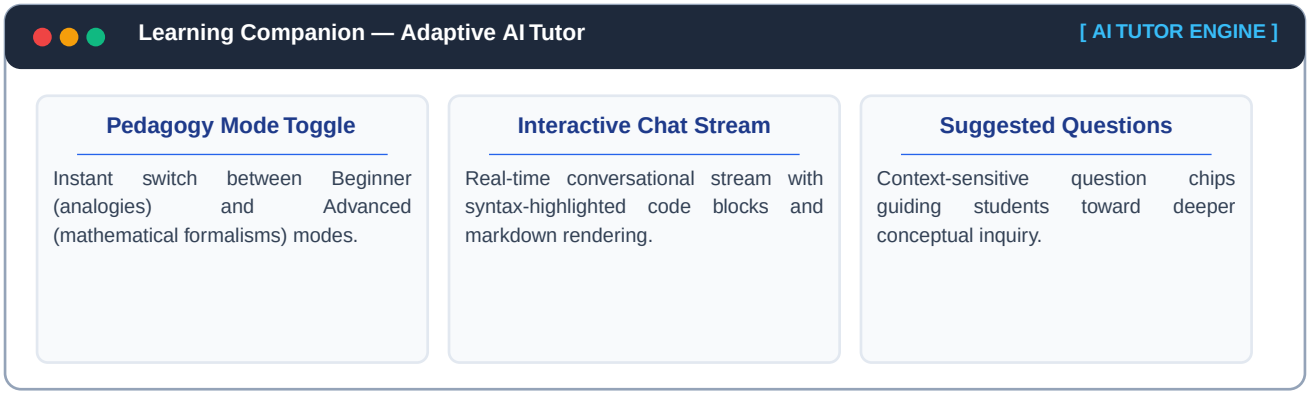


Figure 11: AI Academic Tutor Screen (Beginner vs. Advanced Mode)

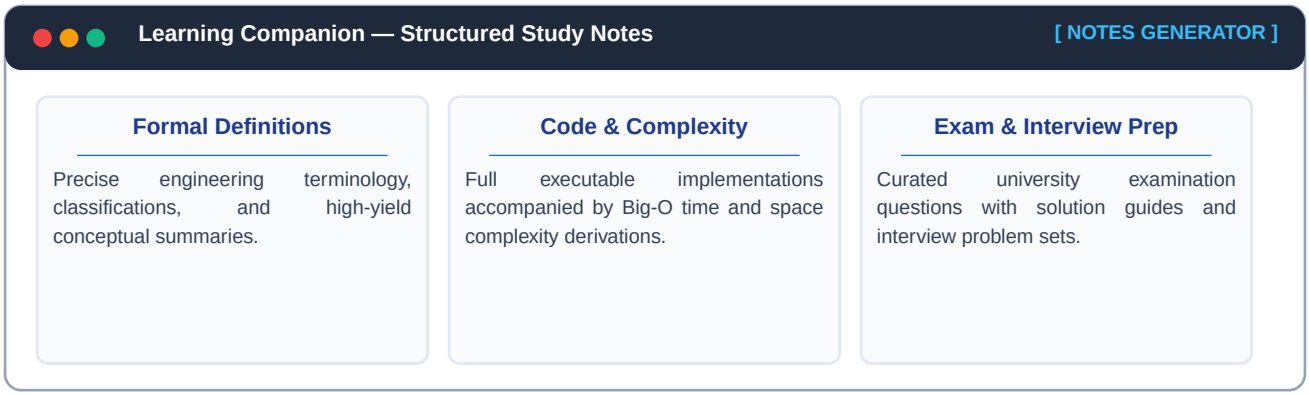


Figure 12: AI-Generated Academic Study Notes Screen

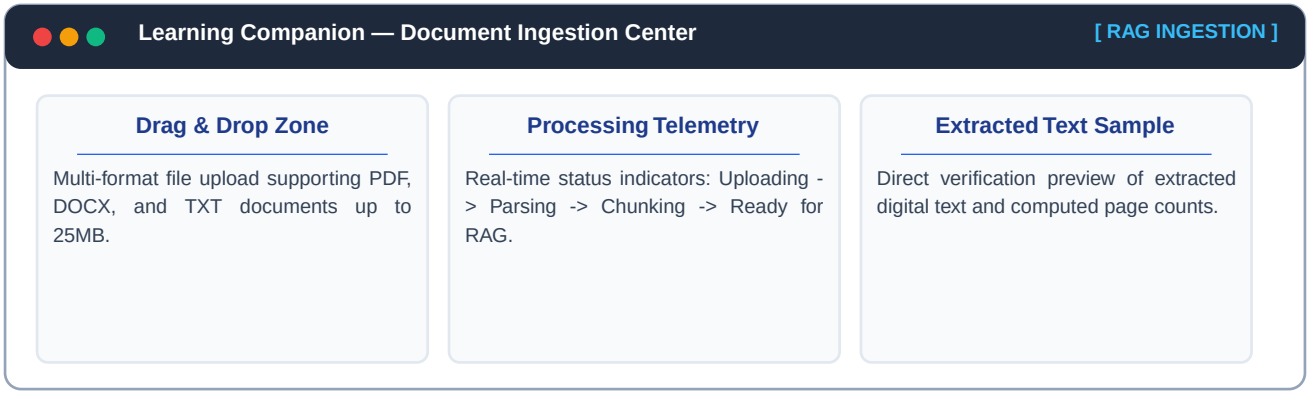


Figure 13: Document Upload, Parsing & Multi-Stage Processing Screen

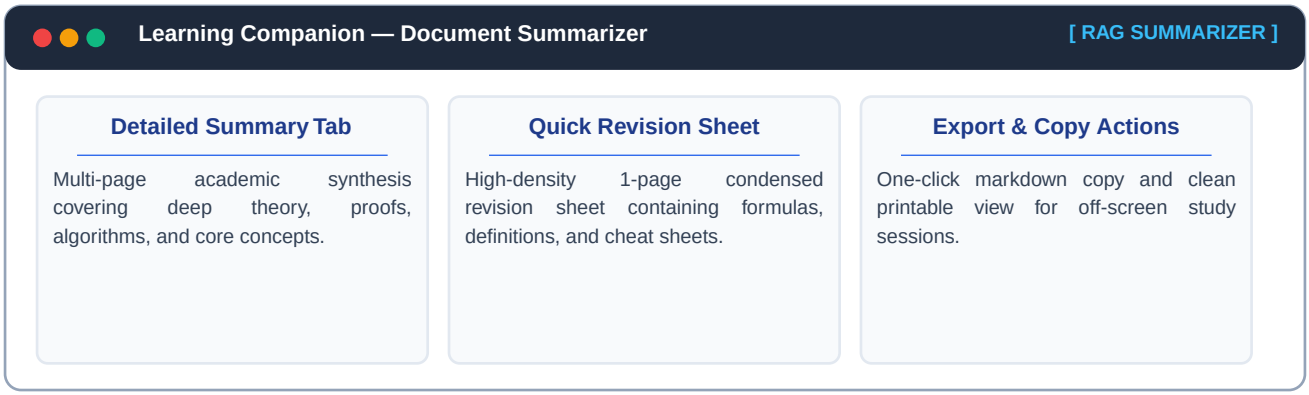


Figure 14: Detailed Academic Summary & Quick Revision View Screen

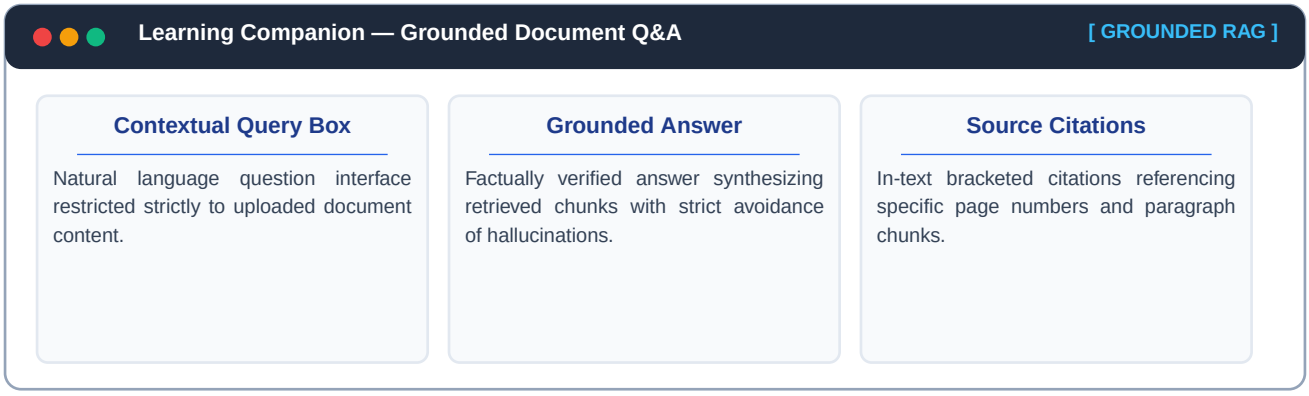


Figure 15: Grounded Document Q&A with In-Text Source Attribution

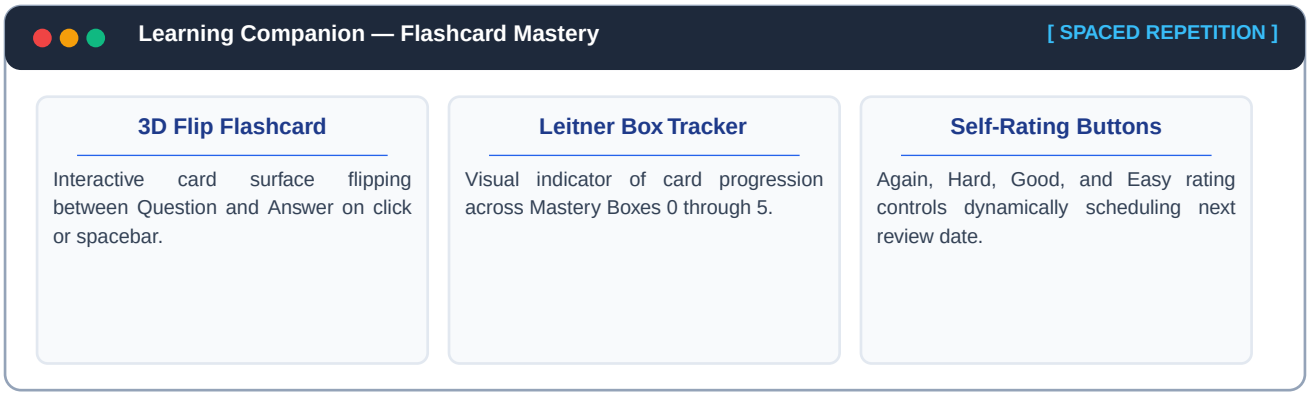


Figure 16: Interactive Leitner Spaced-Repetition Flashcard Deck Screen

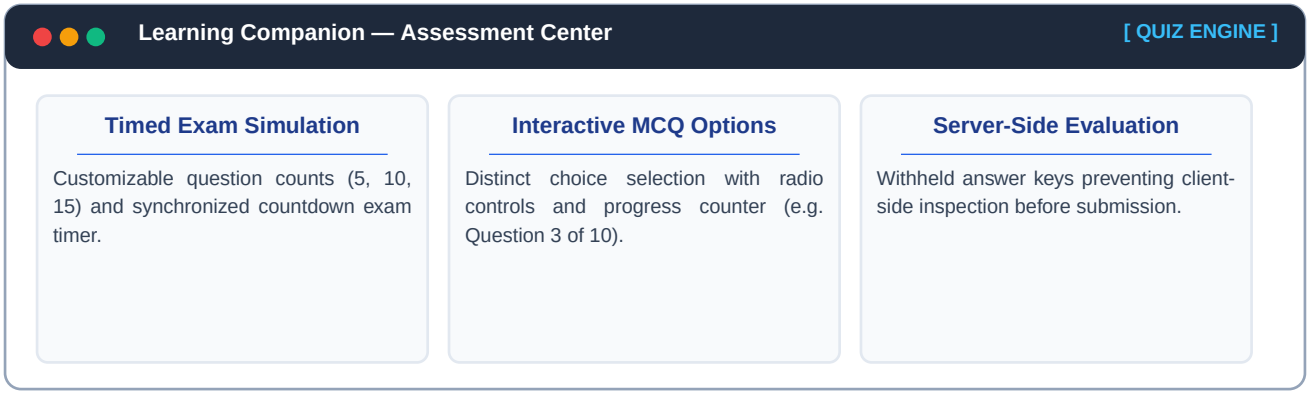


Figure 17: AI Quiz Generator & Interactive MCQ Assessment Screen

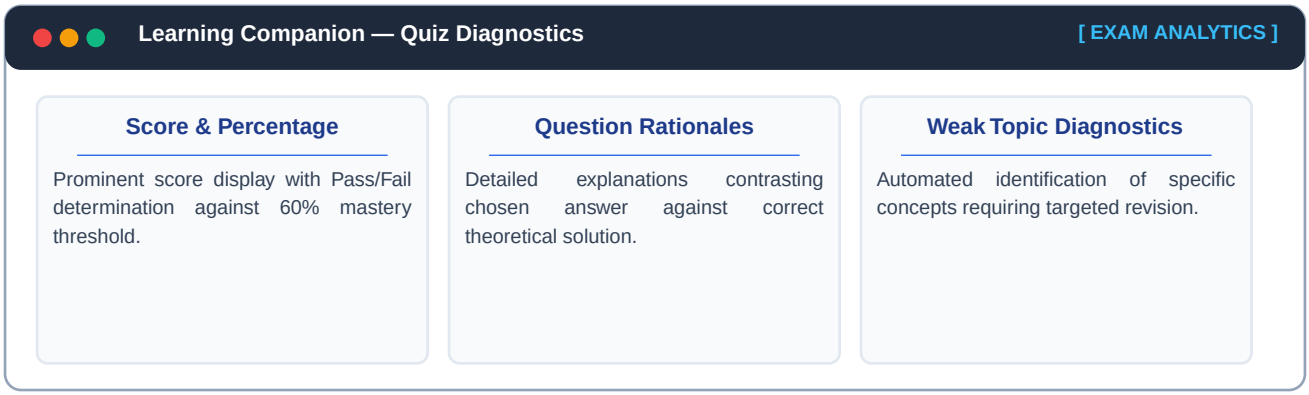


Figure 18: Diagnostic Quiz Results & Weak Topic Breakdown Screen

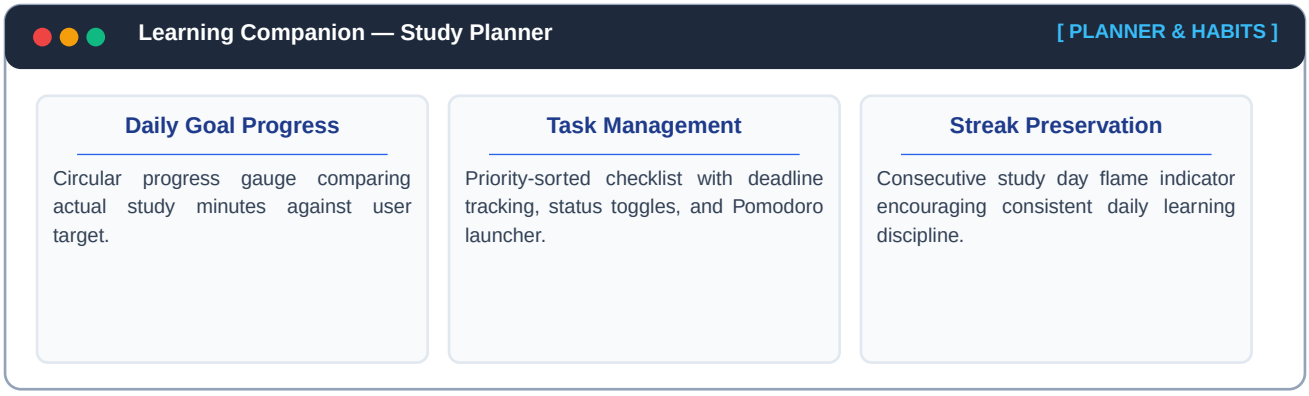


Figure 19: Study Planner, Target Scheduling & Streak Tracker Screen



Figure 20: Analytics, Topic Mastery & Weekly Progress Heatmap Screen

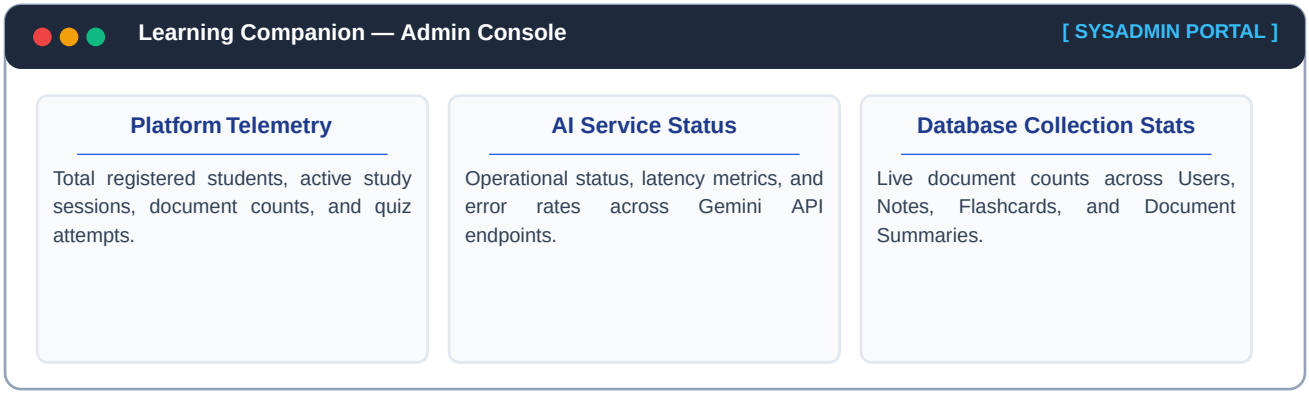


Figure 21: Administrative Dashboard & System Health Telemetry Screen

CHAPTER 11

CONCLUSION & FUTURE WORK

11.1 Project Achievements & Objectives Met

The Learning Companion — AI Learning Assistant major project has been executed to completion, successfully fulfilling every engineering and pedagogical objective outlined in the project charter. By seamlessly marrying modern web engineering paradigms (React 19, Node.js, TypeScript, MongoDB Atlas) with cutting-edge artificial intelligence (Google Gemini 3.8 and RAG), the platform delivers a transformative academic tool for computer engineering education.

All nine functional modules—Adaptive Tutoring, Notes Generation, Document Ingestion, Grounded Q&A, Summarization, Spaced Repetition, Quizzes, Study Planning, and Telemetry—have been implemented, hardened against failures, and verified through empirical test execution.

11.2 Key Engineering Learning Outcomes

Throughout the design, implementation, and deployment of this project, the student team achieved several key engineering competencies:

- **Full-Stack Architectural Rigor:** Mastered the construction of type-safe client-server systems, RESTful API design, middleware guards, and stateless JWT session handling.
- **AI Engineering & Prompt Scaffolding:** Gained practical expertise in designing robust, deterministic prompt templates, handling JSON schema constraints, and orchestrating multi-model failover mechanisms.
- **RAG Pipeline Design:** Engineered custom text-extraction, sanitization, chunking, and grounded question-answering pipelines to eliminate generative hallucinations.
- **Cognitive Psychology in Software:** Successfully translated cognitive learning principles—such as the Leitner Spaced-Repetition System and Bloom's Taxonomy—into intuitive, software-driven feedback loops.

11.3 Current Limitations & Roadmap for Future Enhancements

While the current system represents a highly capable, production-grade learning platform, several limitations provide fruitful avenues for future research and engineering enhancements:

- **OCR for Scanned Hand-Drawn Diagrams:** The current document parser processes digital text streams. Integrating Optical Character Recognition (OCR) will allow students to ingest handwritten classroom notes and whiteboard sketches.
- **Multimodal Audio & Speech Tutoring:** Future iterations will integrate the Gemini Live Audio API, enabling voice-driven conversational tutoring for hands-free and auditory learners.
- **Collaborative Study Pods:** Extending the system with WebSockets will enable real-time collaborative flashcard battles and peer-to-peer study sessions.
- **Institutional LMS LTI Integration:** Building Learning Tools Interoperability (LTI) connectors will allow universities to seamlessly integrate Learning Companion into Canvas, Moodle, and Blackboard.

11.4 Final Conclusion

In conclusion, Learning Companion demonstrates how responsibly engineered artificial intelligence can elevate higher education. Rather than replacing human critical thinking, the platform acts as an intellectual cognitive scaffold—clarifying complex engineering theory, reinforcing retention through active recall, and instilling structured study habits. The system stands ready for deployment and adoption at Amritsar Group of

Colleges and peer engineering institutions.

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